INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab 08: Tiny Block Cipher II *April 23, 2026*

Please do the following exercises by your own using C/C++ programming language and bitwise operators.

1. Deciphering

In this section you will design and implement the required functions to decipher with the tiny block cipher (TBC). Remember that the block size will be $|M| = 16$ bits, i.e. two characters; the key size will be $|K| = 16$ bits, and the number of rounds $r = 3$. In the last session of the laboratory you encipher a block of plaintext using the Algorithm 1.

Algoritmo 1: Tiny block cipher

Input : M a block of plaintext of 16 bits, $K = k_0k_1$ a pseudorandom key, where $|k_i|$=8 bits
Output: C a block of ciphertext

for $i \leftarrow 0$ **to** 2 **do**
  $M \leftarrow M \oplus K_i$
  $M \leftarrow S(M)$ `//substitution`
end
$C \leftarrow M$
return C

1. Write **by hand** the pseudocode (using the notation that we gave in this course) to decipher a block of ciphertext C.
2. Do and example by hand to see if your deciphering algorithm is correct, i.e, first encipher two characters to obtain the ciphertext C and then decipher to recover the plaintext M i.e. the two characters.
3. Implement the pseudocode that you designed. Use functions and remember that K_0, K_1, K_2 are generated using the key expansion and this is an internal part of the block cipher.
4. Test your deciphering function with different pair of letters.

2. Adding a permutation to the TBC

Do the following exercises by your own, using C/C++ programming language and bitwise operations.

1. Design and implement a function that receives as a parameter a permutation P of length 8, and an unsigned char variable s . Your function must use the permutation P to permute the bits of s i.e. your function must return the result of apply this permutation to s, i.e. $P(s)$
2. Modify your TBC previously implemented to add a permutation. Thus the algorithm to encipher must be as Algorithm 2

Algoritmo 2: Tiny block cipher

Input : M a block of plaintext, $K = k_0k_1k_2k_3$ a pseudorandom key, where $|k_i|$=8 bits
Output: C a block of ciphertext

```
for i ← 0 to 2 do
    M ← M ⊕ k_i
    M ← S(M) //substitution
    M ← P(M) //permutation
end
C ← M
return C
```

3. Modify the function to decipher (that you implemente previously) to consider that a permutation was used in the enciphering process.

3. Products

Every student must write her/his own document to present results. Your document must include

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Pseudocode and source code of your function to decipher (Section 1)
3. Pseudocode and source code of your function to decipher with the permutation included. (Section 2)
4. Screenshots of your source code running for section 1 and section 2.